

NAME

landlock_restrict_self – enforce a Landlock ruleset

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <linux/landlock.h> /* Definition of LANDLOCK_* constants */
#include <sys/syscall.h> /* Definition of SYS_* constants */

int syscall(SYS_landlock_restrict_self, int ruleset_fd,
            uint32_t flags);
```

DESCRIPTION

Once a Landlock ruleset is populated with the desired rules, the **landlock_restrict_self()** system call enables enforcing this ruleset on the calling thread. See **landlock(7)** for a global overview.

A thread can be restricted with multiple rulesets that are then composed together to form the thread's Landlock domain. This can be seen as a stack of rulesets but it is implemented in a more efficient way. A domain can only be updated in such a way that the constraints of each past and future composed rulesets will restrict the thread and its future children for their entire life. It is then possible to gradually enforce tailored access control policies with multiple independent rulesets coming from different sources (e.g., init system configuration, user session policy, built-in application policy). However, most applications should only need one call to **landlock_restrict_self()** and they should avoid arbitrary numbers of such calls because of the composed rulesets limit. Instead, developers are encouraged to build a tailored ruleset thanks to multiple calls to **landlock_add_rule(2)**.

In order to enforce a ruleset, either the caller must have the **CAP_SYS_ADMIN** capability in its user namespace, or the thread must already have the *no_new_privs* bit set. As for **seccomp(2)**, this avoids scenarios where unprivileged processes can affect the behavior of privileged children (e.g., because of set-user-ID binaries). If that bit was not already set by an ancestor of this thread, the thread must make the following call:

```
prctl(PR_SET_NO_NEW_PRIVS, 1, 0, 0, 0);
```

ruleset_fd is a Landlock ruleset file descriptor obtained with **landlock_create_ruleset(2)** and fully populated with a set of calls to **landlock_add_rule(2)**.

flags must be 0.

RETURN VALUE

On success, **landlock_restrict_self()** returns 0.

ERRORS

landlock_restrict_self() can fail for the following reasons:

EOPNOTSUPP

Landlock is supported by the kernel but disabled at boot time.

EINVAL

flags is not 0.

EBADF

ruleset_fd is not a file descriptor for the current thread.

EBADFD

ruleset_fd is not a ruleset file descriptor.

EPERM

ruleset_fd has no read access to the underlying ruleset, or the calling thread is not running with *no_new_privs*, or it doesn't have the **CAP_SYS_ADMIN** in its user namespace.

E2BIG The maximum number of composed rulesets is reached for the calling thread. This limit is currently 64.

STANDARDS

Linux.

HISTORY

Linux 5.13.

EXAMPLES

See **landlock(7)**.

SEE ALSO

landlock_create_ruleset(2), **landlock_add_rule(2)**, **landlock(7)**